Groups 1 and 7, gases in the atmosphere and reactivity

Mark scheme and worked solutions for 30 questions: 10 multiple choice, 20 structured.

1 · Part 1

Paper 1 & 2 | Group 1 (alkali metals) – lithium, sodium and potassium | [1]

Answer: B

Worked solution

The general reaction is metal + water → metal hydroxide + hydrogen. Lithium hydroxide, sodium hydroxide and potassium hydroxide are all soluble alkalis, so the water becomes alkaline. The metals differ in how vigorously they react, but they all make the same kinds of product, which is why they are grouped together.

2 · Part 1

Paper 2 only | Group 1 (alkali metals) – lithium, sodium and potassium | [1]

Answer: D

Worked solution

All three atoms have one outer electron, which they lose when they react. The more occupied shells an atom has, the further the outer electron is from the nucleus and the weaker the attraction holding it, so it is lost more easily. Y has four occupied shells, X has three and Z has two. Order of reactivity, most reactive first: Y, X, Z.

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3 · Part 1

Paper 1 & 2 | Group 1 (alkali metals) – lithium, sodium and potassium | [6]

Mark scheme

(a) **[2]**

Any two (1 each):

- the metal floats (on the water) (1)
- fizzing OR bubbles (of gas) (1)
- the metal moves around on the surface (1)
- the metal gets smaller OR disappears (1)
- the indicator turns blue OR purple (1)

REJECT melts into a ball OR catches fire OR lilac flame (these are not seen for all three metals)

(b) **[2]**

- correct product formulae NaOH and H_2 (1)
- fully correct: $2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)$, balanced with state symbols (aq) and (g) (1)

ALLOW second mark only if both products are correct

(c) **[2]**

- they all react in a similar way OR show similar observations OR have similar chemical properties (1)
- they all form the same kind of products, an alkaline metal hydroxide (solution) and hydrogen (1)

ALLOW they all have one electron in their outer shell, as an alternative for the second mark

Worked solution

(a) Lithium, sodium and potassium are all less dense than water, so all three float. Each one fizzes as hydrogen is given off, moves about on the surface and gradually disappears. The hydroxide formed is an alkali, so universal indicator turns blue or purple. Only sodium and potassium melt into a ball, and only potassium catches fire with a lilac flame, so these do not count.

(b) Sodium + water → sodium hydroxide + hydrogen. Hydrogen is H_2, so two sodium atoms are needed to make one H_2 molecule: $2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)$. Sodium hydroxide dissolves, so it is (aq); hydrogen is a gas, (g).

(c) Elements in a family have similar chemical properties. All three metals react with water in the same way and make the same type of products, a metal hydroxide and hydrogen. This is evidence that they belong in the same group.

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4 · Part 1

Paper 1 & 2 | Group 1 (alkali metals) – lithium, sodium and potassium | [6]

Mark scheme

(a) **[2]**

- potassium > sodium > lithium (1)
- potassium turns dull fastest OR lithium stays shiny longest, so potassium reacts fastest with (the oxygen in) air (1)

(b) **[1]**

- to stop them reacting with air (oxygen) OR water (vapour) (1)

ALLOW to keep air and water away from them

(c) **[1]**

- Li_2O (1)

(d) **[2]**

Any two (1 each):

- reacts more vigorously than potassium OR violently OR explodes (1)
- very rapid fizzing OR bubbles (1)
- disappears very quickly (1)
- catches fire OR flame (1)

ALLOW melts (into a ball) OR moves (very) rapidly on the surface

IGNORE floats OR sinks

IGNORE forms an alkaline solution (not an observation unless an indicator colour is stated)

Worked solution

(a) The dull layer forms because the metal reacts with oxygen in the air. The faster the surface turns dull, the faster the reaction and the more reactive the metal. Potassium goes dull in seconds, sodium in about a minute and lithium only after several minutes, so the order is potassium, sodium, lithium.

(b) Oil keeps air and moisture away from the metals, so they do not react while stored.

(c) Lithium forms Li^+ ions and oxide ions are O^{2-}. Two Li^+ balance one O^{2-}, so the formula is Li_2O.

(d) Reactivity increases down Group 1, so rubidium is more reactive than potassium. It would react even more violently, with very rapid fizzing, may catch fire or explode, and would disappear very quickly.

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5 · Part 1

Paper 1 & 2 | Group 1 (alkali metals) – lithium, sodium and potassium | [6]

Mark scheme

(a) **[1]**

- melting point decreases (down the group) (1)

(b) **[1]**

- any value from 30 to 60 °C (1)

REJECT values outside this range OR no unit

(c) **[2]**

- liquid (1)
- 35 °C is above the melting point of caesium (28 °C) (1)

ALLOW second mark for "it has melted because the temperature is higher than 28 °C"

(d) **[2]**

- the density generally increases down the group OR lithium is the least dense and caesium the most dense (1)
- but potassium (0.86 g/cm^3) is less dense than sodium (0.97 g/cm^3), so the statement is not fully supported OR there is an exception (1)

Worked solution

(a) Reading down the table, the melting points fall: 181, 98, 63 and 28 °C.

(b) Rubidium is between potassium (63 °C) and caesium (28 °C), so its melting point should lie between these values, roughly 40–50 °C. Its actual melting point is about 39 °C.

(c) Caesium melts at 28 °C. At 35 °C it is above its melting point, so it would be a liquid.

(d) The densities rise overall from 0.53 g/cm^3 for lithium to 1.93 g/cm^3 for caesium, which supports the general idea. However, potassium is less dense than sodium, so the trend is not regular and the statement is only partly true.

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6 · Part 1

Paper 2 only | Group 1 (alkali metals) – lithium, sodium and potassium | [5]

Mark scheme

(a) **[1]**

- 1 OR one (1)

(b) **[3]**

- both atoms have one electron in their outer shell, which is lost when they react (1)
- the outer electron in potassium is further from the nucleus OR potassium has more (occupied) electron shells (1)
- so the outer electron in potassium is less strongly attracted (to the nucleus) and is lost more easily (1)

ALLOW more shielding by inner electrons in potassium as part of the second or third marking point

(c) **[1]**

- 2.8 (1)

Worked solution

(a) Every Group 1 atom has one electron in its outer shell.

(b) Lithium is 2.1 and potassium is 2.8.8.1. Both react by losing their single outer electron. In potassium this electron is in the fourth shell, much further from the nucleus than lithium's outer electron in the second shell, and there are more inner electrons between it and the nucleus. The attraction is weaker, so potassium loses its outer electron more easily and reacts more vigorously.

(c) A sodium atom is 2.8.1. It loses its outer electron to form Na^+, leaving 2.8.

7 · Part 1

Paper 1 & 2 | Group 7 (halogens) – chlorine, bromine and iodine | [1]

Answer: A

Worked solution

Down Group 7 the halogens get darker and their melting and boiling points increase: chlorine is a yellow-green gas, bromine a red-brown liquid and iodine a grey-black solid.

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8 · Part 1

Paper 2 only

Group 7 (halogens) – chlorine, bromine and iodine

[1]

Answer: C

Worked solution

Halogen atoms have seven outer electrons and react by gaining one more. Going down the group, atoms have more shells, so the outer shell is further from the nucleus and more shielded. An incoming electron is attracted less strongly, so reactivity decreases down the group. Of the four atoms listed, chlorine is highest in the group, so it gains an electron most easily.

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9 · Part 1 Paper 1 & 2 Group 7 (halogens) – chlorine, bromine and iodine [6]

Mark scheme

(a) **[1]**

- brown (1)

ALLOW orange to brown OR dark brown

REJECT stays orange OR purple OR black (solid)

(b) **[3]**

- chlorine displaces bromine and iodine (from potassium bromide and potassium iodide) (1)
- bromine displaces iodine but not chlorine (1)
- iodine does not displace chlorine or bromine (1)

ALLOW "a more reactive halogen displaces a less reactive halogen from its compounds" in place of any one of the three points

(c) **[1]**

- there would be no reaction OR chlorine cannot displace itself OR it is the same halogen (1)

(d) **[1]**

- the pure halogens are toxic OR harmful OR corrosive, so dilute solutions are safer (1)

ALLOW chlorine is a toxic gas OR bromine is a corrosive (toxic) liquid

ALLOW solutions mix more easily with the halide solutions OR a solution is easier to measure out OR add drop by drop

ALLOW iodine is a solid that is almost insoluble in water, so a solution is needed for the ions to react

Worked solution

(a) Bromine is more reactive than iodine, so it displaces iodine from potassium iodide: the mixture turns brown as iodine is formed.

(b) A more reactive halogen displaces a less reactive halogen from a solution of its halide; the colour change shows that a new halogen has been formed. Chlorine water gives orange (bromine) with bromide and brown (iodine) with iodide, so chlorine is more reactive than both. Bromine water turns iodide brown but leaves chloride unchanged, so bromine is between chlorine and iodine. Iodine leaves both chloride and bromide unchanged, so it is the least reactive.

(c) Mixing chlorine with chloride ions cannot produce a different halogen, so no change would be seen.

(d) Pure chlorine is a toxic gas and pure bromine is a toxic, corrosive liquid. Dilute aqueous solutions are much safer to handle in a laboratory.

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10 · Part 1 Paper 1 & 2 Group 7 (halogens) – chlorine, bromine and iodine [6]

Mark scheme

(a) **[1]**

- boiling point increases (down the group) (1)

(b) **[2]**

- gas (1)
- 70 °C is above the boiling point of bromine (59 °C) (1)

(c) **[2]**

- solid (1)
- black OR very dark grey (1)

ALLOW darker than iodine

REJECT any light colour

(d) **[1]**

- any value from 200 to 400 °C (1)

REJECT values at or below 184 °C OR no unit

Worked solution

(a) The boiling points rise from −188 °C for fluorine to 184 °C for iodine.

(b) Bromine boils at 59 °C, so at 70 °C it is above its boiling point and is a gas.

(c) Melting points increase down the group and iodine is already a solid (melting point 114 °C), so astatine will be a solid at room temperature. The colours of the halogens get darker down the group, and iodine is grey-black, so astatine is expected to be black.

(d) The boiling point increases by roughly 90–150 °C from one halogen to the next, so astatine should boil at roughly 184 + 125 ≈ 300 °C. Any sensible value above that of iodine, in the range 200–400 °C, is accepted.

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11 · Part 1 Paper 2 only Group 7 (halogens) – chlorine, bromine and iodine [6]

Mark scheme

(a) **[1]**

- 2.8.7 (1)

(b) **[3]**

- both atoms have seven outer electrons and gain one electron when they react (1)
- the outer shell of a chlorine atom is closer to the nucleus OR chlorine has fewer (occupied) electron shells (1)
- so the incoming electron is attracted more strongly (by the nucleus) and is gained more easily (1)

ALLOW less shielding in chlorine as part of the second or third marking point

REJECT chlorine loses electrons more easily (max 1 for the part)

(c) **[2]**

- a reaction takes place OR iodine displaces astatine (1)
- because iodine is more reactive than astatine (1)

Worked solution

(a) Chlorine has atomic number 17: 2 + 8 + 7 electrons, so 2.8.7.

(b) Halogen atoms react by gaining one electron to complete their outer shell. Bromine atoms have more occupied shells than chlorine atoms, so bromine's outer shell is further from the nucleus and more shielded by inner electrons. A chlorine atom therefore attracts the extra electron more strongly and gains it more easily, so chlorine is more reactive.

(c) Reactivity decreases down Group 7 and astatine is below iodine, so iodine is more reactive. A more reactive halogen displaces a less reactive one from a solution of its halide, so iodine displaces astatine from sodium astatide.

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12 · Part 1 Paper 1 & 2 Group 7 (halogens) – chlorine, bromine and iodine [5]

Mark scheme

(a) **[1]**

- loss of electrons (1)

(b) **[2]**

- iodide ions lose electrons (to form iodine), so they are oxidised (1)
- chlorine (molecules) gain electrons (to form chloride ions), so chlorine is reduced (1)

REJECT "oxidation and reduction happen at the same time" with no electron transfer described

(c) **[2]**

- chlorine OR Cl_2 (1)
- it gains electrons OR it takes electrons from the iodide ions OR it oxidises the iodide ions (1)

REJECT sodium iodide OR iodine as the oxidising agent (0 marks for the part)

Worked solution

(a) Oxidation is the loss of electrons; reduction is the gain of electrons.

(b) In sodium iodide the iodine is present as iodide ions, I^-. These ions each lose an electron and join in pairs to form iodine molecules, so they are oxidised. Each chlorine molecule gains two electrons to form two chloride ions, Cl^-, so chlorine is reduced. Because oxidation and reduction happen together, it is a redox reaction.

(c) An oxidising agent causes another substance to be oxidised by taking electrons from it. Chlorine takes electrons from the iodide ions, so chlorine is the oxidising agent (and is itself reduced).

13 · Part 1 Paper 1 & 2 Gases in the atmosphere [1]

Answer: A

Worked solution

Nitrogen is by far the main gas in air (about four-fifths), oxygen is about one-fifth, argon is just under 1% and carbon dioxide is only a very small fraction, about 0.04%.

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14 · Part 1 Paper 1 & 2 Gases in the atmosphere [1]

Answer: C

Worked solution

Thermal decomposition means breaking down a compound using heat alone. Metal carbonates such as copper(II) carbonate decompose to give the metal oxide and carbon dioxide: $CuCO_3(s) \rightarrow CuO(s) + CO_2(g)$. No other reactant is involved.

15 · Part 1 Paper 1 & 2 Gases in the atmosphere [1]

Answer: D

Worked solution

Oxygen exists as O_2 molecules. When sulfur burns with a blue flame, each sulfur atom combines with one O_2 molecule to make one molecule of sulfur dioxide, a gas at room temperature: $S(s) + O_2(g) \rightarrow SO_2(g)$.

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16 · Part 1

Paper 1 & 2 Gases in the atmosphere [7]

Mark scheme

(a) **[2]**

- the iron reacts with OR uses up the oxygen in the trapped air (as it rusts) (1)
- so the volume of gas in the tube decreases and water rises to take its place (1)

ALLOW the pressure inside the tube decreases, so water is pushed up

(b) **[2]**

- volume of oxygen = 50.0 − 39.5 = 10.5 cm^3 (1)
- percentage = 10.5 ÷ 50.0 × 100 = 21.0% (1)

ALLOW 21%

ECF correct percentage from an incorrect volume of oxygen

(c) **[1]**

- so that all the oxygen has reacted OR the reaction is complete (1)

(d) **[1]**

- iron wool has a larger surface area, so it reacts faster OR removes all the oxygen in the time (1)

(e) **[1]**

- 20.8% (1)

ECF (answer to (b) + 20.6 + 20.8) ÷ 3

Worked solution

(a) Rusting uses up oxygen from the trapped air. As the oxygen is removed, the volume of gas inside the tube gets smaller, so water from the beaker moves up the tube to fill the space.

(b) The decrease in volume equals the volume of oxygen: 50.0 − 39.5 = 10.5 cm^3. Percentage of oxygen = 10.5 ÷ 50.0 × 100 = 21.0%. Check: 21.0% of 50.0 cm^3 is 10.5 cm^3.

(c) Rusting is slow. If the final reading were taken while the water was still rising, some oxygen would remain and the percentage would be too low.

(d) Iron wool has many thin strands, so much more iron surface is in contact with the air and the oxygen is used up much faster than by one nail.

(e) Mean = (21.0 + 20.6 + 20.8) ÷ 3 = 62.4 ÷ 3 = 20.8%.

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17 · Part 1 Paper 1 & 2 Gases in the atmosphere [7]

Mark scheme

(a) **[1]**

- so that all the oxygen comes into contact with the hot copper and reacts (1)

(b) **[2]**

- decrease in volume = 90.0 − 71.2 = 18.8 cm^3 (1)
- percentage = 18.8 ÷ 90.0 × 100 = 20.9% (1)

ALLOW 21% OR 20.89%

ECF correct percentage from an incorrect decrease in volume

(c) **[2]**

- gases expand when heated OR the volume of a gas depends on its temperature (1)
- so the volumes must all be measured at the same (room) temperature as the starting volume OR otherwise the readings would be too high (1)

(d) **[2]**

- all the copper would react before all the oxygen had been removed (1)
- so the decrease in volume would be smaller and her percentage of oxygen would be too low (1)

Worked solution

(a) Only the oxygen that touches the hot copper can react with it. Passing the air over the copper many times makes sure all the oxygen meets the copper and is removed.

(b) The volume stops changing at 71.2 cm^3, so all the oxygen has been removed. Volume of oxygen = 90.0 − 71.2 = 18.8 cm^3. Percentage = 18.8 ÷ 90.0 × 100 = 20.9%. Check: 20.9% of 90.0 cm^3 is 18.8 cm^3.

(c) Hot gas takes up a larger volume than the same gas at room temperature. The starting volume was measured at room temperature, so every reading must also be taken at room temperature for the decrease to be caused only by the loss of oxygen.

(d) If there is not enough copper, it all turns into copper(II) oxide while some oxygen is still left in the syringes. Less oxygen is removed, so the volume decrease is too small and the calculated percentage of oxygen is lower than the true value.

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18 · Part 1 Paper 1 & 2 Gases in the atmosphere [6]

Mark scheme

(a) **[2]**

- bright white light OR bright white flame (1)
- white solid OR white powder OR white ash forms (1)

ALLOW white smoke

(b) **[2]**

- MgO as the product (1)
- $2Mg(s) + O_2(g) \rightarrow 2MgO(s)$, balanced with state symbol (s) (1)

(c) **[2]**

- water (1)
- hydrogen gains oxygen OR combines with oxygen (1)

Worked solution

(a) Magnesium burns with a very bright white flame and leaves a white powder, magnesium oxide.

(b) Magnesium ions are Mg^{2+} and oxide ions are O^{2-}, so magnesium oxide is MgO. One O_2 molecule gives two oxygen atoms, so two Mg are needed: $2Mg(s) + O_2(g) \rightarrow 2MgO(s)$.

(c) Hydrogen combines with oxygen to form water, H_2O. Oxidation is the gain of oxygen, and hydrogen gains oxygen, so the hydrogen is oxidised.

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19 · Part 1 Paper 1 & 2 Gases in the atmosphere [5]

Mark scheme

(a) **[1]**

- green to black (1)
- both colours needed

(b) **[1]**

- copper(II) carbonate, zinc carbonate, magnesium carbonate, sodium carbonate (1)

(c) **[2]**

Any two (1 each):

- the same mass of carbonate (1)
- the same volume of limewater (1)
- the same concentration of limewater OR limewater from the same bottle (1)
- the same strength of heating OR the same Bunsen burner flame OR the same distance from the flame (1)
- the same size of test tube OR delivery tube (1)
- the same particle size of carbonate (1)

(d) **[1]**

- to stop cold limewater being sucked back into the hot test tube (which could crack it) (1)

ALLOW to prevent suck-back

Worked solution

(a) Copper(II) carbonate is a green powder. When heated it decomposes into copper(II) oxide, which is black, and carbon dioxide.

(b) The shorter the time for the limewater to turn milky, the faster carbon dioxide is given off and the more easily the carbonate decomposes. Copper(II) carbonate is quickest (25 s), then zinc carbonate (55 s), then magnesium carbonate (140 s); sodium carbonate did not decompose at all in 600 s.

(c) To compare the carbonates fairly, only the type of carbonate should change. Keep the mass of carbonate, the volume and concentration of limewater and the heating the same.

(d) When heating stops, the gas in the test tube cools and contracts. If the delivery tube is still in the limewater, the limewater can be sucked back into the hot tube and crack it.

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20 · Part 1 Paper 1 & 2 Gases in the atmosphere [6]

Mark scheme

(a) **[2]**

- the concentration increased (from 317 ppm to 414 ppm) (1)
- the rate of increase got faster OR it increased by a larger amount in each 20-year period (22, 31, 44 ppm) (1)

(b) **[2]**

- increase = 414 − 317 = 97 ppm (1)
- percentage increase = 97 ÷ 317 × 100 = 30.6% (1)

ALLOW 31%

ECF correct percentage from an incorrect increase

(c) **[2]**

- carbon dioxide is a greenhouse gas OR it absorbs OR traps heat (radiated from the Earth) (1)
- so the average temperature of the Earth (atmosphere) increases OR causes global warming, which changes the climate (1)

Worked solution

(a) The concentration rose in every 20-year period, from 317 ppm in 1960 to 414 ppm in 2020. The increases were 22, 31 and 44 ppm, so the rise was getting faster.

(b) Increase = 414 − 317 = 97 ppm. Percentage increase = 97 ÷ 317 × 100 = 30.6%. Check: 317 × 1.306 ≈ 414.

(c) Carbon dioxide is a greenhouse gas: it absorbs heat energy given off by the Earth's surface and keeps it in the atmosphere. More carbon dioxide traps more heat, so the average temperature rises, which may change weather patterns and the climate.

21 · Part 1 Paper 1 & 2 Reactivity series [1]

Answer: B

Worked solution

The reactivity series, most reactive first, is: potassium, sodium, lithium, calcium, magnesium, aluminium, zinc, iron, copper, silver, gold. The four metals in B appear in this order.

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22 · Part 1 Paper 1 & 2 Reactivity series [1]

Answer: D

Worked solution

A metal displaces another metal from a solution of its salt only if it is more reactive. Magnesium is above iron in the reactivity series, so magnesium + iron(II) sulfate → magnesium sulfate + iron.

23 · Part 1 Paper 1 & 2 Reactivity series [1]

Answer: A

Worked solution

Aluminium gains oxygen and its atoms lose electrons, so aluminium is oxidised and acts as the reducing agent. Iron(III) oxide loses oxygen and its iron ions gain electrons to form iron atoms, so iron(III) oxide is reduced and acts as the oxidising agent.

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24 · Part 1 Paper 1 & 2 Reactivity series [8]

Mark scheme

(a) **[2]**

- magnesium > zinc > iron (1)
- magnesium fizzes most vigorously AND gives the largest temperature rise; iron fizzes least AND gives the smallest temperature rise (1)

ALLOW second mark for reference to either the fizzing or the temperature rise for all three metals

(b) **[2]**

- $ZnCl_2$ and H_2 as the products (1)
- $Zn(s) + 2HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$, with 2HCl and state symbols (aq) and (g) (1)

ALLOW second mark only if both products are correct

(c) **[4]**

Indicative points, any four (1 each):

- use the same volume of acid, measured with a measuring cylinder (1)
- use acid of the same concentration (1)
- use the same mass of each metal (1)
- use the same form of each metal, e.g. all powders or pieces of the same size, so the surface area is the same (1)
- start with the acid at the same temperature (1)
- measure the starting temperature and the highest temperature with a thermometer to find the temperature rise (1)
- observe the rate of bubbling OR measure the volume of gas given off in a fixed time (1)
- repeat for each metal and calculate a mean (1)

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Worked solution

(a) The more reactive the metal, the faster it reacts with acid, so the faster it fizzes and the more heat it gives out in the time. Magnesium reacts most vigorously with the biggest temperature rise (14.5 °C), then zinc (4.0 °C), then iron (1.5 °C).

(b) Zinc + hydrochloric acid → zinc chloride + hydrogen. Zinc ions are Zn^{2+}, so zinc chloride is $ZnCl_2$, which needs 2HCl: $Zn(s) + 2HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$.

(c) Only the metal should change. Put the same measured volume of acid of the same concentration into each test tube and record its starting temperature. Add the same mass of each metal in the same form (for example, all as powder). Watch how fast it fizzes and record the highest temperature, then work out the temperature rise. A faster fizz and a bigger temperature rise mean a more reactive metal.

25 · Part 1 Paper 1 & 2 Reactivity series [6]

Mark scheme

(a) **[2]**

- calcium hydroxide (1)
- hydrogen (1)

(b) **[2]**

- correct formulae: $Mg + H_2O \rightarrow MgO + H_2$ (1)
- correct state symbols: $Mg(s) + H_2O(g) \rightarrow MgO(s) + H_2(g)$ (1)

REJECT $H_2O(l)$ for the second mark (steam is a gas)

(c) **[2]**

- calcium > magnesium > copper (1)
- calcium reacts with cold water, magnesium needs steam to react quickly, and copper does not react with water or steam (1)

Worked solution

(a) A reactive metal + water → metal hydroxide + hydrogen, so calcium forms calcium hydroxide and hydrogen.

(b) With steam, magnesium forms magnesium oxide and hydrogen gas: $Mg(s) + H_2O(g) \rightarrow MgO(s) + H_2(g)$. Steam is water in the gas state, so it is $H_2O(g)$.

(c) The more easily a metal reacts with water, the more reactive it is. Calcium reacts readily with cold water, magnesium only reacts quickly with steam, and copper does not react at all, so the order is calcium, magnesium, copper.

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26 · Part 1 Paper 1 & 2 Reactivity series [7]

Mark scheme

(a) **[2]**

- T > R > W (1)
- T displaces both R and W; R displaces only W; W displaces neither (1)

(b) **[2]**

Any two (1 each):

- the blue colour of the solution fades OR the solution becomes colourless (1)
- a pink-brown OR red-brown OR brown solid forms (on the zinc) (1)
- the zinc gets smaller OR dissolves (1)
- the mixture gets warmer (1)

REJECT fizzing OR bubbles

(c) **[1]**

- $ZnSO_4(aq) + Cu(s)$ (1)

(d) **[2]**

- zinc atoms lose electrons (to form zinc ions), so zinc is oxidised (1)
- copper(II) ions gain electrons (to form copper atoms), so copper ions are reduced (1)

Worked solution

(a) A metal displaces a less reactive metal from a solution of its salt. T reacts with both other nitrates, so it is the most reactive. R reacts only with W nitrate, so it is in the middle. W reacts with neither, so it is the least reactive: T, R, W.

(b) Zinc is more reactive than copper, so it displaces copper. The blue copper(II) ions are used up, so the blue colour fades, and pink-brown copper metal forms on the zinc, which slowly dissolves. The reaction is exothermic, so the solution warms up.

(c) $Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$.

(d) Each zinc atom loses two electrons to become a zinc ion, Zn^{2+}, so zinc is oxidised. Each copper(II) ion, Cu^{2+}, gains two electrons to become a copper atom, so it is reduced. Oxidation and reduction happen at the same time, so it is a redox reaction.

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27 · Part 1

Paper 1 & 2 | Reactivity series | [6]

Mark scheme

(a) **[2]**

- MgO and Cu as the products (1)
- Mg(s) + CuO(s) → MgO(s) + Cu(s), with state symbols (s) for both products (1)

(b) **[2]**

- copper(II) oxide OR CuO (1)
- it loses oxygen (1)

(c) **[2]**

- no reaction (1)
- iron is less reactive than zinc, so it cannot remove the oxygen from zinc oxide (1)

Worked solution

(a) Magnesium is more reactive than copper, so it takes the oxygen from copper(II) oxide: Mg(s) + CuO(s) → MgO(s) + Cu(s). Both products are solids.

(b) Reduction is loss of oxygen. Copper(II) oxide loses its oxygen to become copper, so copper(II) oxide is reduced (magnesium gains oxygen and is oxidised).

(c) A metal only takes oxygen from the oxide of a less reactive metal. Zinc is above iron in the reactivity series, so iron cannot reduce zinc oxide and there is no reaction.

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28 · Part 1 Paper 1 & 2 Reactivity series [6]

Mark scheme

(a) **[2]**

- boiling removes dissolved air OR oxygen from the water (1)
- the oil stops air OR oxygen from dissolving back into the water (1)

(b) **[1]**

- to absorb water (vapour) from the air OR to keep the air dry (1)

(c) **[2]**

- water and oxygen (air) (1)
- tube 1, with both water and oxygen, rusts; tube 2 (no oxygen) and tube 3 (no water) do not rust (1)

(d) **[1]**

- salt makes iron rust faster OR more (1)

Worked solution

(a) Water normally contains some dissolved air. Boiling drives the dissolved air out, and the layer of oil stops fresh air dissolving in again, so the nail in tube 2 has water but no oxygen.

(b) Anhydrous calcium chloride absorbs water, so the air in tube 3 is dry: the nail has oxygen but no water.

(c) Only tube 1 has both water and oxygen and only tube 1 (of the first three) rusts. Without oxygen (tube 2) or without water (tube 3) there is no rust, so both water and oxygen are needed.

(d) Tube 4 is the same as tube 1 except that the water contains salt, and it has much more rust, so salt speeds up rusting.

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29 · Part 1 Paper 1 & 2 Reactivity series [8]

Mark scheme

(a) **[1]**

- the paint is a barrier that stops oxygen (air) and water reaching the iron (1)

(b) **[4]**

Indicative points, any four (1 each):

- while it is complete, the zinc coating is a barrier that keeps oxygen (air) and water away from the steel (1)
- zinc is more reactive than iron (1)
- when scratched, the zinc reacts OR corrodes instead of the iron (1)
- zinc loses electrons OR is oxidised in preference to iron (1)
- this is sacrificial protection (1)

(c) **[1]**

- the magnesium corrodes OR reacts OR is used up (as it protects the steel) (1)

(d) **[2]**

- copper is less reactive than iron (1)
- so copper would not corrode in place of the iron, and the iron would still rust (1)

ALLOW the iron would rust faster

Worked solution

(a) Rusting needs both oxygen and water. A layer of paint covers the iron so neither can reach it.

(b) A complete zinc layer works like paint: it keeps air and water off the steel. If the coating is scratched, the steel is exposed, but zinc is more reactive than iron. The zinc reacts with the oxygen and water instead of the iron, losing electrons in preference to it, so the iron does not rust. The zinc is sacrificed to protect the iron.

(c) The magnesium blocks protect the hull by reacting instead of the iron. As they do this the magnesium is gradually used up, so the blocks must be replaced.

(d) Sacrificial protection only works if the attached metal is more reactive than iron. Copper is less reactive than iron, so the iron would react instead of the copper and the hull would still rust.

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30 · Part 1 Paper 1 & 2 Reactivity series [7]

Mark scheme

(a) **[3]**

- M is between iron and copper (in the reactivity series) (1)
- M does not displace iron, so M is less reactive than iron (1)
- M displaces copper, so M is more reactive than copper (1)

ALLOW M reacts with dilute acid but copper does not, as an alternative for the third point

(b) **[2]**

- MSO4 (1)
- MCl2 (1)

(c) **[2]**

- yes, it would react (1)
- M is more reactive than copper, so it can remove the oxygen from copper(II) oxide (1)

ECF from the position of M given in (a)

Worked solution

(a) M cannot displace iron from iron(II) sulfate, so M is less reactive than iron. M does displace copper from copper(II) sulfate, so M is more reactive than copper. Its slow reaction with acid and no reaction with cold water also fit a metal just below iron. So M is between iron and copper.

(b) Sulfate ions are SO_4^{2-} and chloride ions are Cl^-. One M^{2+} balances one SO_4^{2-}, giving MSO4. One M^{2+} needs two Cl^-, giving MCl2.

(c) A more reactive metal takes oxygen from the oxide of a less reactive metal. M is more reactive than copper, so M + copper(II) oxide → M oxide + copper.

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